

$$11.12.13.i4. \rightarrow$$

$$\begin{aligned} & \frac{1}{6} m_e^{-2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} + \frac{1}{16\pi^2} \left(-\frac{1}{1296} m_e^{-2} (972 s_\gamma^2 c_\gamma^2 \bar{y}_d^{r13} \bar{y}_d^{12p} y_d^{r14} y_d^{i1p} + s_\gamma^2 (81 \bar{y}_d^{pr} \bar{y}_d^{i2i3} \right. \\ & \quad (y_d^{p14} y_d^{i1r} (1 + s_\gamma^2) - 8 c_\gamma^2 y_d^{pr} y_d^{i1i4}) + 3 \bar{y}_d^{i2i3} (2 y_d^{i1i4} (17 g_1^2 + 81 g_2^2 + \\ & \quad 240 g_3^2) - 9 y_d^{p14} \bar{y}_d^{pr} y_u^{i1r} (-1 + c_\gamma^2)) + \bar{y}_d^{p13} (81 \bar{y}_d^{i2r} y_d^{pr} y_d^{i1i4} (1 + s_\gamma^2) + \\ & \quad 27 y_d^{i2r} (-y_d^{i1i4} y_u^{pr} (-1 + c_\gamma^2) + 12 c_\gamma^2 y_d^{p14} y_u^{i1r}) + 2 g_1^2 y_d^{p14} \delta_{i1i2})) + \\ & \quad 2 g_1^2 (5 s_\gamma^2 \bar{y}_d^{i2p} y_d^{i1p} - 13 c_\gamma^2 \bar{y}_d^{i2p} y_u^{i1p}) \delta_{i3i4} \Big) - \\ & \frac{1}{12} \sum p s_\gamma g_1^2 \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) LF_{1,0}[m_d^p] - \\ & \frac{2}{243} \sum p g_1^4 LF_{3,0}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{5}{324} \sum p g_1^4 LF_{4,-1}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{8}{1215} \sum p g_1^4 LF_{5,-2}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{2} s_\gamma \frac{1}{m_e^4} \bar{y}_d^{pr} \bar{y}_d^{i2i3} y_d^{pr} y_d^{i1i4} (-s_{2\gamma} c_\gamma + s_\gamma^3) LF_{1,0}[m_d^r] - \\ & \frac{1}{12} \sum p s_\gamma g_1^2 \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) LF_{1,0}[m_e^p] - \\ & \frac{81}{21} \sum p g_1^4 LF_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{5}{108} \sum p g_1^4 LF_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{8}{405} \sum p g_1^4 LF_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{6} s_\gamma \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} \bar{y}_e^{pr} y_e^{pr} (-s_{2\gamma} c_\gamma + s_\gamma^3) LF_{1,0}[m_e^r] + \\ & \frac{1}{12} s_\gamma \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} (2 \bar{y}_e^{pr} y_e^{pr} (s_{2\gamma} c_\gamma - s_\gamma^3) + \sum p g_1^2 (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma})) LF_{1,0}[m_l^p] - \\ & \frac{1}{81} \sum p g_1^4 LF_{3,0}[m_l^p] \delta_{i1i2} \delta_{i3i4} + \frac{5}{216} \sum p g_1^4 LF_{4,-1}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{4}{405} \sum p g_1^4 LF_{5,-2}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \frac{1}{12} s_\gamma \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} \\ & \quad (6 \bar{y}_d^{pr} y_d^{pr} (-s_{2\gamma} c_\gamma + s_\gamma^3) + 6 c_\gamma \bar{y}_d^{pr} y_u^{pr} (s_{2\gamma} + s_\gamma c_\gamma) + \sum p g_1^2 (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma})) \\ & \quad LF_{1,0}[m_q^p] - \frac{1}{243} \sum p g_1^4 LF_{3,0}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{5}{648} \sum p g_1^4 LF_{4,-1}[m_q^p] \delta_{i1i2} \delta_{i3i4} - \frac{4}{1215} \sum p g_1^4 LF_{5,-2}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{6} \sum p s_\gamma g_1^2 \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) LF_{1,0}[m_u^p] - \\ & \frac{8}{243} \sum p g_1^4 LF_{3,0}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \frac{5}{81} \sum p g_1^4 LF_{4,-1}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{32}{1215} \sum p g_1^4 LF_{5,-2}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{2} s_\gamma c_\gamma \frac{1}{m_e^2} \bar{y}_d^{i2i3} y_d^{i1i4} \bar{y}_d^{pr} y_u^{pr} (s_{2\gamma} + s_\gamma c_\gamma) LF_{1,0}[m_u^r] - \frac{1}{24} s_\gamma \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} \\ & \quad (3 s_{4\gamma} c_\gamma (g_1^2 + g_2^2) + s_\gamma (g_1^2 (-1 + 3 c_{2\gamma}^2) + 3 g_2^2 (-1 + c_{2\gamma}^2))) LF_{1,0}[m_\Phi] + \\ & \frac{1}{24} \frac{1}{m_e^2} s_\gamma^2 (3 s_\gamma^2 \bar{y}_d^{pr} \bar{y}_d^{i2i3} y_d^{p14} y_d^{i1r} + 3 \bar{y}_d^{p13} y_d^{i1i4} (s_\gamma^2 \bar{y}_d^{i2r} y_d^{pr} - c_\gamma^2 \bar{y}_d^{i2r} y_u^{pr}) - \\ & \quad \bar{y}_d^{i2i3} (2 y_d^{i1i4} (g_1^2 + 3 g_2^2) + 3 c_\gamma^2 y_d^{p14} \bar{y}_d^{pr} y_u^{i1r})) LF_{1,1}[m_\Phi] + \frac{1}{216} \\ & \quad (s_\gamma^2 (-108 c_\gamma^2 \bar{y}_d^{r13} \bar{y}_d^{i2p} y_d^{r14} y_d^{i1p} + 9 \bar{y}_d^{i2i3} (y_d^{i1i4} (g_1^2 + 3 g_2^2) + 2 c_\gamma^2 y_d^{p14} \bar{y}_d^{pr} y_u^{i1r}) + \\ & \quad 2 \bar{y}_d^{p13} (9 c_\gamma^2 \bar{y}_d^{i2r} (y_d^{i1i4} y_u^{pr} - 6 y_d^{p14} y_u^{i1r}) - 2 g_1^2 y_d^{p14} \delta_{i1i2})) - \\ & \quad 8 g_1^2 (s_\gamma^2 \bar{y}_d^{i2p} y_d^{i1p} - 2 c_\gamma^2 \bar{y}_d^{i2p} y_u^{i1p}) \delta_{i3i4}) LF_{1,2}[m_\Phi] + \\ & \frac{1}{36} (-9 s_\gamma^4 \bar{y}_d^{r13} \bar{y}_d^{i2p} y_d^{r14} y_d^{i1p} + s_\gamma^2 \bar{y}_d^{p13} y_d^{p14} (9 c_\gamma^2 \bar{y}_d^{i2r} y_u^{i1r} + g_1^2 \delta_{i1i2}) + \\ & \quad g_1^2 (s_\gamma^2 \bar{y}_d^{i2p} y_d^{i1p} - c_\gamma^2 \bar{y}_d^{i2p} y_u^{i1p}) \delta_{i3i4}) LF_{2,1}[m_\Phi] - \\ & \frac{1}{324} g_1^2 (9 s_\gamma^2 \bar{y}_d^{p13} y_d^{p14} \delta_{i1i2} + (9 s_\gamma^2 \bar{y}_d^{i2p} y_d^{i1p} - 9 c_\gamma^2 \bar{y}_d^{i2p} y_u^{i1p} + 4 g_1^2 \delta_{i1i2}) \delta_{i3i4}) \\ & \quad LF_{3,0}[m_\Phi] + \frac{5}{216} g_1^4 LF_{4,-1}[m_\Phi] \delta_{i1i2} \delta_{i3i4} - \frac{4}{405} g_1^4 LF_{5,-2}[m_\Phi] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{81} g_1^4 LF_{3,0}[\bar{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{54} g_1^4 LF_{4,-1}[\bar{\mu}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{405} g_1^4 LF_{5,-2}[\bar{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} LF_{1,1,0}[m_1, m_d^{i3}] - \\ & \frac{1}{108} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} LF_{2,1,-1}[m_1, m_d^{i3}] - \\ & \frac{1}{972} g_1^4 LF_{2,1,0}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{972} g_1^4 LF_{2,2,-1}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{486} g_1^4 LF_{3,1,-1}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{972} g_1^4 LF_{4,1,-2}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{54} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} LF_{1,1,0}[m_1, m_d^{i4}] - \\ & \frac{1}{108} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} LF_{2,1,-1}[m_1, m_d^{i4}] - \frac{1}{972} g_1^4 LF_{2,1,0}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{972} g_1^4 LF_{2,2,-1}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{486} g_1^4 LF_{3,1,-1}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{972} g_1^4 LF_{4,1,-2}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{216} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} LF_{1,1,0}[m_1, m_q^{i1}] - \\ & \frac{1}{432} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} LF_{2,$$